

Paper A-12: Earth Synthesis: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

ORCID: 0009-0001-0556-5516

May 2026

Abstract

This paper closes the Part A Earth-systems sequence. It treats atmosphere, ocean, hydrology, soil, ecology, biodiversity, pollution, agriculture, sustainability, collapse, plate tectonics, and volcanology as different transport settings for the same CDFD/AFL audit variables: flux Φ , constraint C , responsiveness S , and structural memory M_s . The synthesis does not merge these disciplines into one mechanism. It gives a common way to ask what is flowing, what limits it, how the system responds, and what memory keeps shaping later behavior.

1 Series Context

Part I sets the flow–constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV tests whether the notation remains useful when it is moved into Earth systems and other domains.

Notation. In this paper, C names the active Earth-system constraint or capacity burden. Φ is the driving flow, S is the response or buffering capacity, and M_s is retained structural memory. Λ appears only where the Part II Life Number is referenced [2].

2 Part A Integration

The Part A sequence is ordered as a transport ladder. Atmospheric circulation and ocean currents move heat and moisture at planetary scale. Hydrological and soil systems route water, sediment, nutrients, and organic matter. Ecological, biodiversity, pollution, and agricultural papers move into coupled biological and human-modified transport. Sustainability and collapse ask when response capacity is lost. Plate tectonics and volcanology extend the same audit language to slow geophysical flow and abrupt release.

CDFD term	Earth-system synthesis reading	Example proxy
Φ	environmental drive or throughput	heat, runoff, sediment, biomass, magma
C	active constraint	aridity, stratification, toxicity, friction, rigidity
S	response capacity	mixing, buffering, biodiversity, storage, rerouting
M_s	retained memory	hysteresis, degradation, legacy load, thermal storage
Ψ_s	operating ratio	balanced, constrained, overloaded, or locked regime

3 Shared Earth-System Map

4 What Carries Across Domains

The common feature is not material identity. Heat transport, river discharge, ecosystem energy flow, pollution load, tectonic strain, and volcanic pressure are physically different. What carries across Part A is the audit question: does rising flux meet a constraint that cannot relax quickly enough, and does prior load make recovery harder? When that question can be written with real proxies, CDFD/AFL gives a compact measurement frame.

5 Measurement Layer

Each Earth-system paper needs a named dataset or field measurement before the mapping becomes more than a modelling claim. The synthesis layer asks for four items: an observable Φ , an observable C , a response proxy S , and a memory proxy M_s . The failure condition can be flood, stagnation, biodiversity loss, contamination persistence, transport weakening, or another domain-specific state. The release does not turn those candidates into operational forecasts.

6 Current Runtime Diagnostic

The Part E diagnostic run includes Earth-system adapter rows for climate and atmosphere, ocean transport, hydrology, biodiversity, and pollution. The universal cascade stress test checks finite behavior and memory locking in the current CDFD Runtime [4]. The numbers are a reproducibility record for the release, not field validation.

7 Tests That Can Break the Mapping

The Part A synthesis weakens if the same datasets are explained more cleanly by standard domain models, if CDFD/AFL variables cannot be separated from ordinary descriptive labels, or if the proposed operating ratio fails to track overload or recovery across independent Earth-system cases.

8 Conclusion

Part A is strongest when it stays close to measurement. Its contribution is a shared language for flow, constraint, response, and memory across Earth systems. The next step

is not broader analogy; it is narrower data work in each domain.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs` and `Part_E_Synthesis/figures/`.

9 Reference Layer

References

- [1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.
- [2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.
- [3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.
- [4] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfd execution engine, 2026.